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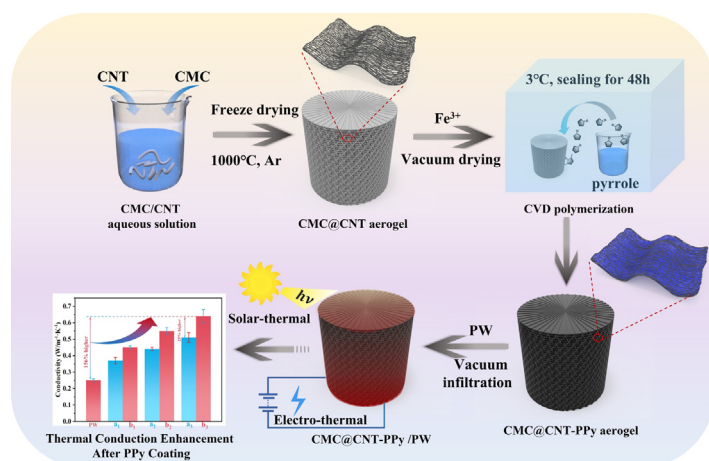
Polypyrrole coated carbon nanotube aerogel composite phase change materials with enhanced thermal conductivity, high solar-/electro-thermal energy conversion and storage

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GRAPHICAL ABSTRACT



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ABSTRACT

Phase change materials (PCMs) have been widely investigated as promising thermal management materials due to their high thermal storage capacity, satisfactory heat transfer rate and multi-responsive energy conversion and storage characteristics. In this work, a shape-stabilized solar-/electro- responsive thermal energy capture and storage system is proposed involving polypyrrole (PPy)-deposited carbon nanotubes (CNT) heterogeneous porous aerogel as a supporting matrix and the paraffin wax (PW) as a PCM. The composite PCMs obtained via integration of PW into aerogel supports present a relatively high thermal storage density of 160.9 J/g and outstanding phase transition stability even after 100 heating-cooling cycles. Furthermore, great enhancement of thermal conductivity ($0.64 \text{ W/m}^{-1}\cdot\text{K}^{-1}$, 2.56 times that of PW) is achieved in the composite PCMs by inducing PPy coating as a binder in the gap between CNTs. The mechanism of heat transport enhancement is explored by molecular dynamics simulation. It

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storage
High thermal conductivity

concludes that the in-situ polymerization of PPy through the vapor deposition method on the CNT aerogels effectively builds additional thermal transfer channels and enhances the heat transport between CNT by coordinating the carbon atom vibration. Herein, this reported stratagem may shed light on preparing composite PCMs with high thermal conductivity and multi-energy utilization functions.

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1. Introduction

Developing efficient energy storage materials to realize energy collection and redistribution is an effective way to overcome uneven energy distribution and alleviate the contradiction between resources and development. Phase change materials (PCMs) obtaining thermal capture and reuse through latent heat absorption and release are considered promising energy storage materials, especially in the thermal energy storage and temperature control fields [1,2]. For instance, PCMs can be applied in waste heat recovery [3], overheating protection of electronic devices [4–6], building insulation [7,8], cold-chain transportation [9], intelligent temperature-control fabrics [10], and infrared stealth [11,12], etc.

Organic PCMs such as paraffin, polyols and fatty acids, etc, with nontoxicity, inexpensive, chemically stable and high thermal storage capacity features are widely investigated in thermal storage materials [2,13]. However, the bottleneck issues of shape instability at phase transition period, low intrinsic thermal conductivity and only response to temperature variations restrict the large-scale application of organic PCMs.

To obtain the demand-satisfied PCMs, materials compositing stratagem with the porous material as the supporting matrix and the PCM as the thermal storage unit are commonly implemented to overcome the shortcomings of organic PCMs. To date, porous supports such as metal foam [14,15], diatomite [16], perlite [17], CNT aerogel [18,19], expanded graphite [20,21], porous polymer [22] and metal organic frameworks [23] are fabricated and used to prevent the shape change of PCMs at the solid–liquid transition period. Among them, carbon-based materials, due to their relatively low density, good heat transfer performance and substantial light absorption property are regarded as a type of ideal [supporting materials](#) utilized in the composite PCMs [24]. Carbon nanotubes as carbon-based supports with high axial thermal conductivity are frequently used in composite PCMs. In most cases, the one-dimensional CNT are always assembled into three-dimensional porous networks, such as CNT foam or CNT aerogel and thus the PCM can be encapsulated stably in the pores by capillary force. Meanwhile, after integrating PCM with the CNT network, the thermal conductivity of composite PCMs can also be improved due to the continuous heat transfer path established by the CNT network. For the composite PCM composed of a CNT network, the improvement of the thermal conductivity is influenced by the amount of CNT and heat transfer efficiency between CNT. High density CNT networks can rapidly improve thermal conductivity of composite PCMs but also significantly reduce its thermal storage density. Thus, the optimization of inter-CNT heat transfer efficiency becomes a decisive factor for further improving the thermal conductivity of composite PCM. The weak inter-CNT force and strong interfacial thermal resistance between the CNTs slow down the transfer rate of phonons between the interfaces and increase the scattering. It is an effective method to improve the inter-CNT force by introducing crosslinking agent. For examples, Chen et al. prepared the CNT/eicosane composite PCMs via colloid aggregation by adding the calcium chloride solution and achieving high energy storage capacity and conductivity of 204.8 J/g and 620.3 S/m [25].

Maleki et al. designed a three-dimensional interconnected porous polystyrene-CNT foam scaffold and the obtained paraffin/polystyrene-CNT composites exhibited enhanced thermal conductivity $0.39 \text{ W m}^{-1} \text{ K}^{-1}$, high PCM loading of 82.5% and latent heat of 119.3 J/g [26]. Hu and co-workers reported a flexible phase change composites consisting of PW and cross-linked styrene butadiene styrene(SBS) and CNT interpenetrating network, the obtained composite PCM displayed enhanced thermal conductivity of $0.39 \text{ W m}^{-1} \text{ K}^{-1}$ and high phase change enthalpy of 147.63 J/g [27]. The addition of these crosslinked materials improves the inter-CNT force, but has a slight enhancement to the thermal conductivity of the composite PCMs. Therefore, it is crucial to choose suitable materials to improve the inter-CNT force and reduce the interfacial thermal resistance of CNT networks. In addition to the enhancement of thermal conductivity, enriching the methods of thermal energy capture of PCM is also an effective way to expand its application. CNT can convert electric energy into heat energy and release it through joule heat, which provides the basis for electro-thermal conversion and storage of CNT-based composite PCMs [28,29]. Meanwhile, the light absorption properties of CNT also make it possible to convert light energy into thermal energy. Herein, on the basis of further improving the thermal conductivity, it is an effective means to obtain high-performance CNT-based composite PCMs with solar-thermal and electro-thermal conversion characteristics.

In this study, the three-dimensional porous aerogel [supporting materials](#) with high thermal conductivity were designed by smartly integrating CNT aerogel skeleton with continuous conducting-polymer coating. CNT and carboxymethyl cellulose (CMC) formed a mixed solution in water, and then CMC/CNT aqueous solution was converted into CNT aerogel (CMC@CNT) via freeze drying and then calcination. Further, the inner and outer surfaces of CMC@CNT aerogel were coated with polypyrrole (PPy) by chemical vapor deposition (CVD). The shape-stabilized composite PCMs (CMC@CNT-X-PPy/PW) were obtained via vacuum infiltrating paraffin wax (PW) into the supporting matrix. The thermal storage capacity, thermal conductivity and solar-/electro- thermal energy conversion and storage properties of the composited PCMs CMC@CNT-X-PPy/PW, were investigated in detail. The molecular dynamics simulation was employed to count the efficacy of in-situ polymerized PPy coatings between CNTs for improving the thermal conductivity of the [supporting material](#).

2. Experimental section

2.1. Materials

The multi-walled carbon nanotubes (with specific surface area $280 \text{ m}^2/\text{g}$, outer diameter 10–15 nm, length about $50 \mu\text{m}$) were purchased from Nanjing XFNANO Materials-Tech Co., Ltd. The carboxy methyl cellulose (CMC, the average molecular weight (Mn) 90,000 and degree of substitution 0.7) and paraffin wax (PW, density of $\sim 0.9 \text{ g/cm}^3$ at 20°C) were obtained from Sinopharm Chemical Reagent Co., Ltd. Chemical reagents including pyrrole ($\sim 99.7\%$) and ethanol ($\sim 99.5\%$) were supplied by Sigma-Aldrich Company

and used without further purification. The ferric chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, 98%) was provided by the Beijing National Medicines Company. Ultra-pure deionized water ($18.2 \text{ M}\Omega \cdot \text{cm}^{-1}$) was gained by a Master system.

2.2. Preparation of three-dimensional porous CNT aerogel

The preparation process of three-dimensional carbon nanotube (CNT) aerogel was carried out as shown in Fig. 1. Firstly, the carboxy methyl cellulose (CMC, 300 mg) was dissolved into 10 ml of deionized water, and then carbon nanotubes (CNTs) (200 mg, 300 mg or 400 mg) were dispersed in above solution. Next, the mixture was magnetically stirred and ultrasonically treated to obtain a stable and evenly dispersed CMC/CNT solution. The ultrasonic processor used the gap pulse mode and the ultrasonic time was 20 min, the power was 600 W. After freeze drying, the three-dimensional CMC/CNT aerogels were obtained. Finally, as-made CMC/CNT aerogels were heated at 1000°C for 1 h with a heating rate of $2^\circ\text{C}/2^\circ\text{C}$ per minute in a tube furnace under an inert atmosphere and thus the three-dimensional porous CNT aerogels were obtained. The prepared porous CNT aerogels were denoted as CMC@CNT-X, where the X represented the added weight amount of CNT and could be 200, 300 or 400.

2.3. Preparation of CNT and polypyrrole heterogeneous porous aerogel

To maintain the pore structure of CMC@CNT-X aerogels, an efficient method of vapor deposition polymerization was used to form continuous polypyrrole coating on the inner and outer surface of CMC@CNT-X aerogel network. In a typically procedure, the CMC@CNT-X aerogels were immersed into 0.04 mol/L $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ solution and then CMC@CNT-X aerogels containing the iron ions were obtained after vacuum drying. Next, the aerogel and pyrrole liquid were both sealed in a container without direct contact, as shown in Fig. 1. Finally, the aforementioned container was placed at 3°C for 48 h, the CNT - polypyrrole (PPy) heterogeneous porous

aerogels were obtained, and they were denoted accordingly as CMC@CNT-X-PPy (X = 200, 300 or 400).

2.4. Preparation of composite phase change materials

The paraffin wax (PW) was encapsulated into CMC@CNT-X and CMC@CNT-X-PPy aerogels by vacuum infiltration. Firstly, the PW was melted at 80°C . Then the aerogels of CMC@CNT-X and CMC@CNT-X-PPy were immersed into melting PW, and kept them under vacuum at 80°C for 8 h to achieve saturated adsorption. After that, the excess PW adhered on the three-dimensional aerogel surface was removed by filter paper, and thus a series of composited phase change materials were obtained. The mass fraction of PW in CMC@CNT-200/PW, CMC@CNT-300/PW and CMC@CNT-400/PW were 89.0%, 87.7% and 82.2%, respectively. The mass fraction of PW in CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW were 85.7%, 83.3% and 80.9%, respectively. The calculation formula was shown in Eq. (1):

$$\omega = \left(1 - \frac{m_a}{m_c}\right) \times 100\% \quad (1)$$

where ω was the mass fraction of PW in composite PCMs, m_a represented the mass of aerogel supports, and m_c was the mass of the composite PCMs.

2.5. Characterizations

The morphology and microstructures of the materials were characterized by field emission scanning electron microscopy (SEM, Hitachi SU8000, Japan) and transmission electron microscope (TEM, JEM-100CX, Japan). The chemical structures of the samples were measured on a Fourier Transform Infrared Spectra (FT-IR, Nicolet 6700 by the KBr pellet technique). The crystallization of the samples was characterized by an X-ray diffractometer (M21X, Cu $\text{K}\alpha$ radiation, $\lambda = 1.541 \text{ \AA}$). The thermal stability was evaluated using a synchronous thermal analyzer (TGA, Netzsch STA449F3, Germany) at a heating rate of $10^\circ\text{C}/\text{min}$ under a flowing

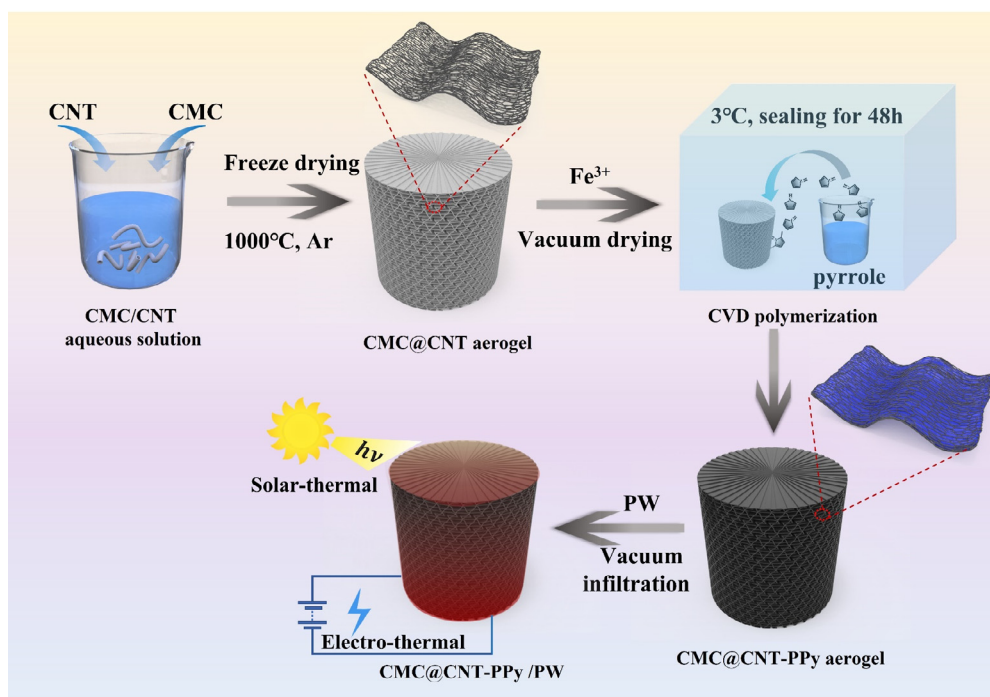


Fig. 1. Schematic description of the preparation of the composite PCM (CMC@CNT-PPy/PW) and its solar-/electro-thermal properties.

nitrogen atmosphere. The pore characters of supports were measured by N₂ adsorption–desorption isotherms (APSP 2460 adsorption analyzer) and automatic mercury injection apparatus (USA Mack corporation Auto Pore IV 9510). The element composition and binding energy were studied by X-ray photoelectron spectra (XPS, escalab 250Xi). Differential scanning calorimeter (DSC, Q2000) measurement was used to characterize the thermophysical properties of the composite PCMs with the heating and cooling rate of 10 K min^{−1} in a nitrogen atmosphere. The heat transfer efficiency of the samples was characterized by an infrared thermal imaging spectrometer (FLUKE TiX580, USA). The thermal conductivities of the composite PCMs were measured using a constant thermal analyzer at room temperature (LFA 467, the samples were pressed into disc-shaped with a diameter of 12.7 mm and a thickness of 1 mm. Then, the pressed standard samples were caught in the disc of the analyzer to complete the test. Each sample was tested three times to ensure the accuracy and reliability of the results.). UV–vis–NIR absorption spectra were examined by the Shimadzu UV–vis–NIR spectrophotometer. The simulated solar irradiation was supplied by Xenon lamp (Solar Edge700 BF, 300 W) equipped with a standard AM1.5 filter. The electro-thermal test was conducted by a direct-current power supply. The temperature data of the electro-thermal samples were recorded by the fiber temperature sensor and the data were collected by an R2100 data acquisition system.

2.6. Molecular dynamic simulations

Molecular dynamics is used to explore the mechanism of heat transfer enhancement. Since the overlapping interface between CNTs dominates the thermal conductivity of large-scale CNT materials such as CNT aerogels, the interfacial thermal conductance of CNT is focused on, and the detailed methods of molecular dynamics simulations were provided in the [supporting information](#) section.

2.7. Solar-/electro-thermal energy storage efficiency evaluation

The solar-thermal energy storage efficiency (η_1) under the simulated solar irradiation was calculated by the Eq. (2) [30]:

$$\eta_1 = \frac{Q_t}{Q} = \frac{m\Delta H}{PA\tau} \quad (2)$$

where Q_t was the stored thermal energy, Q was the received light energy, m represented the mass of the sample, P was the intensity of the light, A ($\sim 1.26 \text{ cm}^2$) was the exposed area of the sample to light, and τ was the time of illumination corresponding to the starting temperature to the ending temperature of the phase transition.

The electro-thermal energy storage efficiency (η_2) under dc power was calculated by the formula (3) [31]:

$$\eta_2 = \frac{Q_t}{Q} = \frac{m\Delta H}{UIt} \quad (3)$$

where Q_t was the stored thermal energy, Q was the received electric energy, m represented the mass of the sample, U was the operating voltage, I presented the current, and t was the time from the starting temperature to the ending temperature of the phase transition.

3. Results and discussion

3.1. Characterizations of aerogel supports and their composite PCMs

The detailed preparation process of three-dimensional porous aerogel supports and their composite PCMs are schematically described in Fig. 1. The CMC@CNT aerogel is obtained after freeze

drying and calcination. The absorbed ferric ion (Fe^{3+}) on the CMC@CNT aerogel can react in situ with volatile pyrrole monomer in a confined space by chemical vapor deposition. At low temperatures, pyrrole polymerizes gently and forms a continuous polypyrrole (PPy) coating on the inner and outer surfaces of CNT. Then, the shape-stabilized composite PCM (CMC@CNT-PPy/PW) with solar-thermal and electro-thermal energy conversion and storage characteristics is constructed by integrating the PW with CMC@CNT-PPy aerogel supports via vacuum infiltration.

The morphologies and structure of the CNT based three-dimensional porous aerogel supports are characterized by SEM. Fig. 2a, and 2b present the morphologies of CMC@CNT-200 and CMC@CNT-200-PPy, respectively. Apparently, they all show lamellar morphology with significant gaps between layers on the micron scale. With the increase addition amount of CNT, for the CMC@CNT-300 and CMC@CNT-300-PPy samples (Fig. 2c and 2d), a prominent bridge structure appears between CNT layers, and a pore-like structure is formed between the layers. Further, as the amount of CNT is 400 mg, the CMC@CNT-400 and CMC@CNT-400-PPy aerogels (Fig. 2e and 2f) both show a continuous regular macropore structure of about 20 μm in size, which is connected with each other. The formation of pores should be caused by the squeezing of CNT by frozen ice crystals as the content of CNT increases within a limited volume range. By comparing Fig. 2a to Fig. 2f, it can be found that the in-situ deposition of PPy coating on the surface of aerogels does not affect the original macropore channel structure of CNT aerogels, which is beneficial to maintaining the porosity of the aerogels. After further magnifying the images, it can be seen that before the deposition of PPy, the pore wall of aerogels is formed by overlapping of CNT which are entangled with each other, and the morphology of each CNT is visible, as shown in the insets of Fig. 2a, 2c and 2e. By comparison, the insets of Fig. 2b, 2d and 2f present a denser appearance, the CNT are joined to each other by a glue-like substance and the cracks between the nanotubes are reduced. In addition, the morphology of one-dimensional CNT can still be observed, and the surface of CNT becomes smoother. These indicate that the pyrrole is successfully in-situ polymerized at the junctions of the CNT and the cracks between the tubes. The EDS mapping detection is performed for the CMC@CNT-300-PPy aerogel (Fig. 2g), and the results are shown in Fig. 2h. The uniform distribution of elements confirms the even distribution of PPy coating in the CMC@CNT-300 aerogel skeleton. The TEM images of CMC@CNT-300 and CMC@CNT-300-PPy samples are displayed in Fig. 2i and 2j, respectively. As can be seen from Fig. 2i, the meandering CNT in the CMC@CNT-300 sample are interlaced and overlapped with each other, and the contact area between CNT is small. However, for the CMC@CNT-300-PPy aerogel supports, the interlaced CNT are covered with a thin PPy film and the spaces between the tubes are filled with the film, which links the network of CNT together. This demonstrates that a thin uniform PPy coating is formed on the surface of CNT aerogels. After loading the PW into aerogel supports, the composite PCMs CMC@CNT-X/PW and CMC@CNT-X-PPy/PW are obtained. Fig. 2k reveals that the PW infiltrates into the CMC@CNT-300 supports and fully occupies the pore space of the aerogel, indicating the PW is restricted steadily by the CNT networks. Fig. 2l presents that the lamellar channels of the CMC@CNT-300-PPy are also filled with PW. The results indicate that the PPy coating on the CNT aerogels does not significantly affect the loading of PW in three-dimensional porous supports and thus ensures a similar PW load amount and thermal storage density between the composite PCMs with and without the PPy coating.

X-ray diffraction (XRD) patterns and Raman spectra are presented to analyze crystal structure of the composites. CMC@CNT-200, CMC@CNT-300 and CMC@CNT-400 aerogel samples show a similar diffraction peak at 2θ of 26.4° and 42.5° , corresponding to

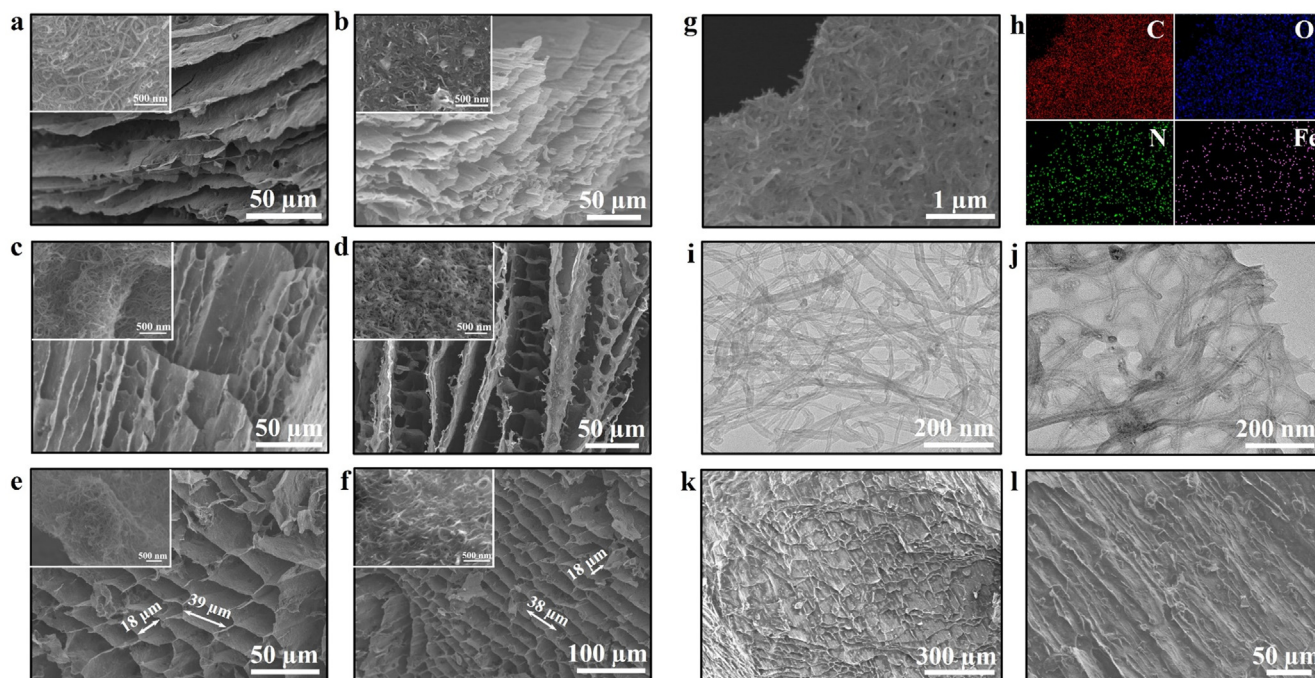


Fig. 2. SEM images of the aerogels: (a) CMC@CNT-200, (b) CMC@CNT-200-PPy, (c) CMC@CNT-300, (d) CMC@CNT-300-PPy, (e) CMC@CNT-400, (f-g) CMC@CNT-400-PPy (the insets of Fig. 2a to 2f are magnifying SEM image of the corresponding samples); (h) Elemental maps of CMC@CNT-400-PPy aerogel corresponding to Fig. 2g; TEM images of (i) CMC@CNT-400, (j) CMC@CNT-400-PPy; SEM images of the composite PCMs: (k) CMC@CNT-300/PW, (l) CMC@CNT-300-PPy/PW.

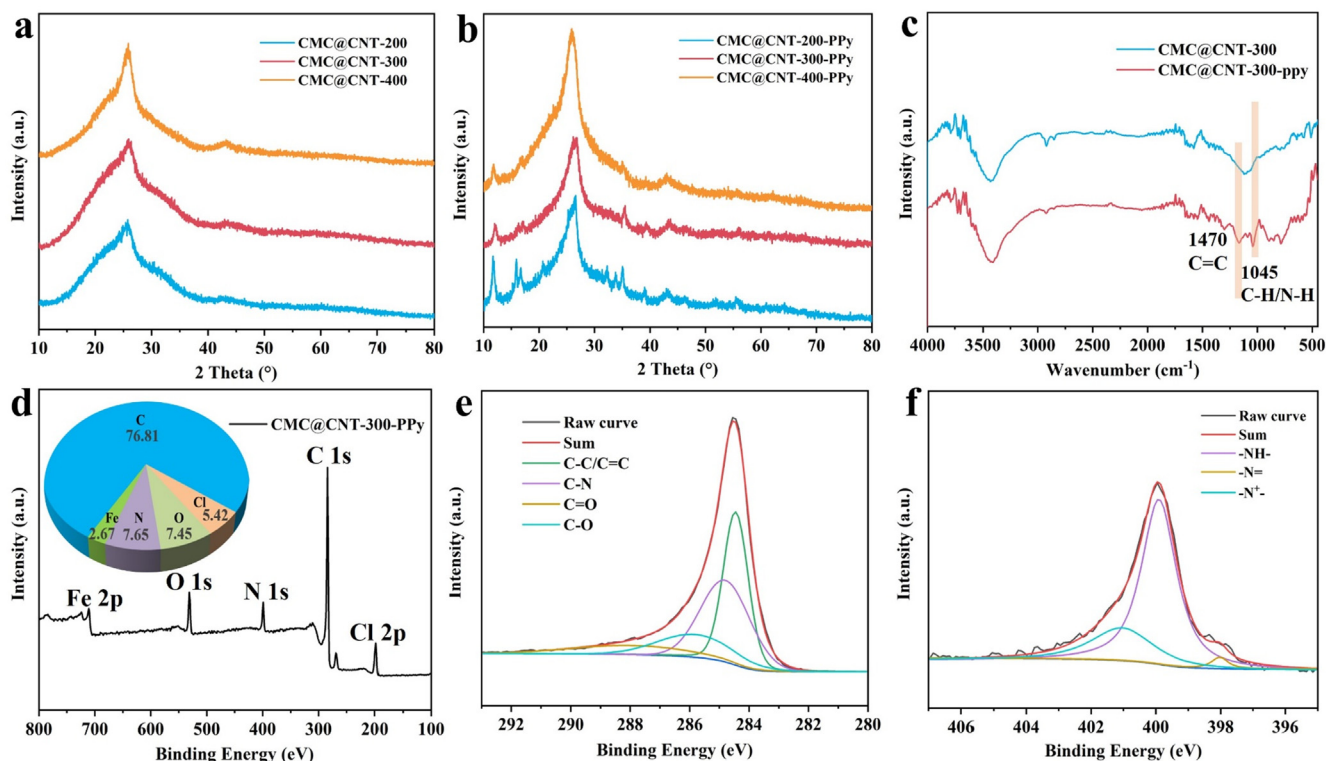


Fig. 3. XRD patterns of (a) CMC@CNT-X and (b) CMC@CNT-X-PPy; (c) FT-IR spectra of CMC@CNT-300 and CMC@CNT-300-PPy; (d) XPS survey spectrum of CMC@CNT-300-PPy (the inset is the pie chart of elements content), XPS wide-scan spectra of (e) C 1s and (f) N 1s.

(002) and (100) lattice plane of CNT, respectively (Fig. 3a) [32]. Raman spectra (Figure S1) of CMC@CNT-X and CMC@CNT-X-PPy also show clear D and G characteristic peaks at about 1346 and 1582 cm^{-1} , originating from amorphous carbon in the tube walls

and the tangential vibrations of the perfect graphite lattice of CNT, respectively [31]. After in-situ polymerization of PPy, the aerogels of CMC@CNT-200-PPy, CMC@CNT-300-PPy and CMC@CNT-400-PPy also appear to have similar XRD characteristic

peaks (Fig. 3b). The diffraction peak at 11.9° , 16.8° , 26.7° , 35.1° and 55.6° are assigned to the (110), (200), (310), (211) and (521) lattice planes of β -FeOOH which is formed by hydrolysis of ferric chloride [33]. Moreover, the characteristic peaks of CNT still exist in Fig. 3b, indicating the PPY coating physically adheres to the aerogel surface. Further, the composite PCM of CMC@CNT-X-PPy/PW presents a similar characteristic peak as PW and CNT (Figure S2), which means no new phase formation after the integration of PW into CMC@CNT-X-PPy aerogel supports. The FT-IR analyses of CMC@CNT-X and CMC@CNT-X-PPy aerogels are performed to verify the chemical composition of the samples (Figure S3). The peak appearing at 3400 cm^{-1} is assigned to the O–H stretching vibration, and the peak around 2900 cm^{-1} is ascribed to the C–H stretching vibration [34]. Besides, the peak at around 1100 cm^{-1} corresponds to the C–C stretching vibration of CNT [35]. By comparing the spectra of CMC@CNT-300 and CMC@CNT-300-PPy samples (Fig. 3c, Figure S3), the peak at about 1470 cm^{-1} and 1045 cm^{-1} are ascribed to the C=C stretching vibration and in-plane deformation of C–H bond and N–H bond of pyrrole ring, respectively [34]. This well demonstrates the formation of PPY coating on aerogels.

The surface element analysis of CMC@CNT-300-PPy aerogel is investigated by X-ray photoelectron spectroscopy (XPS). As shown in Fig. 3d, a wide binding energy scan XPS spectrum is presented which is composed of Cl 2p (198.6 eV), C 1s (284.5 eV), N 1s (399.7 eV), O 1s (531.5 eV) and Fe 2p (710.8 eV). The element composition and corresponding atomic content of CMC@CNT-300-PPy calculated by the XPS spectrum are displayed with a pie chart (the inset of Fig. 3d). The C 1s XPS spectrum of CMC@CNT-300-PPy is deconvoluted into four bands at about 284.5 eV, 285.0 eV, 286.0 eV and 288.1 eV (Fig. 3e), corresponding to C–C/C=C, C–N, C–O and C=O bonding, respectively [36]. The N 1s XPS spectrum (Fig. 3b) of CMC@CNT-300-PPy aerogel is fitted and shows three main peaks at 398.0 eV, 399.9 eV and 401.7 eV, which are corresponding to the -N= , -NH- and -N^+ components, respectively

[34]. These results confirm the polymerization of PPY coating on the surface of aerogel supports. The successful preparation of CMC@CNT-300-PPy heterogeneous porous aerogel provides the possibility for the thermal transfer enhancement of subsequent composite PCMs.

To analyze the pore structure of aerogel supports, the nitrogen adsorption and desorption curves of CMC@CNT-X and CMC@CNT-X-PPy samples are exhibited in Fig. 4a and 4c, respectively. The curves of CMC@CNT-X samples present explicit hysteresis loops and significant micropore adsorption at low relative pressure, indicating that microporous and mesoporous structures are contained in the samples (Fig. 4a) [37]. Nevertheless, for the CMC@CNT-X-PPy samples, the micropore adsorption at low relative pressure almost vanished after the PPY coating formed in the aerogel. Correspondingly, the pore size distribution curves of CMC@CNT-X and CMC@CNT-X-PPy samples (Fig. 4b and Fig. 4c) also revealed that the micropores are distributed in the CMC@CNT-X and do not appear for CMC@CNT-X-PPy samples. Besides, the specific surface area and pore volume of the CMC@CNT-X and CMC@CNT-X-PPy are calculated and listed in Table 1. Comparatively, it is found that the specific surface area and pore volumes are decreased and the average pore size are increased after the PPY coating is formed in the aerogels. These changes in micro-mesoporous structures indicated that pyrrole is successfully polymerized in and out of the CNT aerogels and fully occupied the micropore space in the aerogel, which may be beneficial to reducing the contact thermal resistance during heat transfer between CNT.

The mercury intrusion method is also used to analyze the macroporous structure of aerogels before and after polypyrrole deposition (Figure S4). The corresponding macropore size distribution curves as shown in Fig. 4e, indicate that the macropore sizes of CMC@CNT-X-PPy are mainly distributed between $1\text{ }\mu\text{m}$ to $10\text{ }\mu\text{m}$. The pore structural parameters (Table S1) calculated by the mercury intrusion method of CMC@CNT-X and CMC@CNT-X-PPy samples present that the macroporous structures of aerogels do not

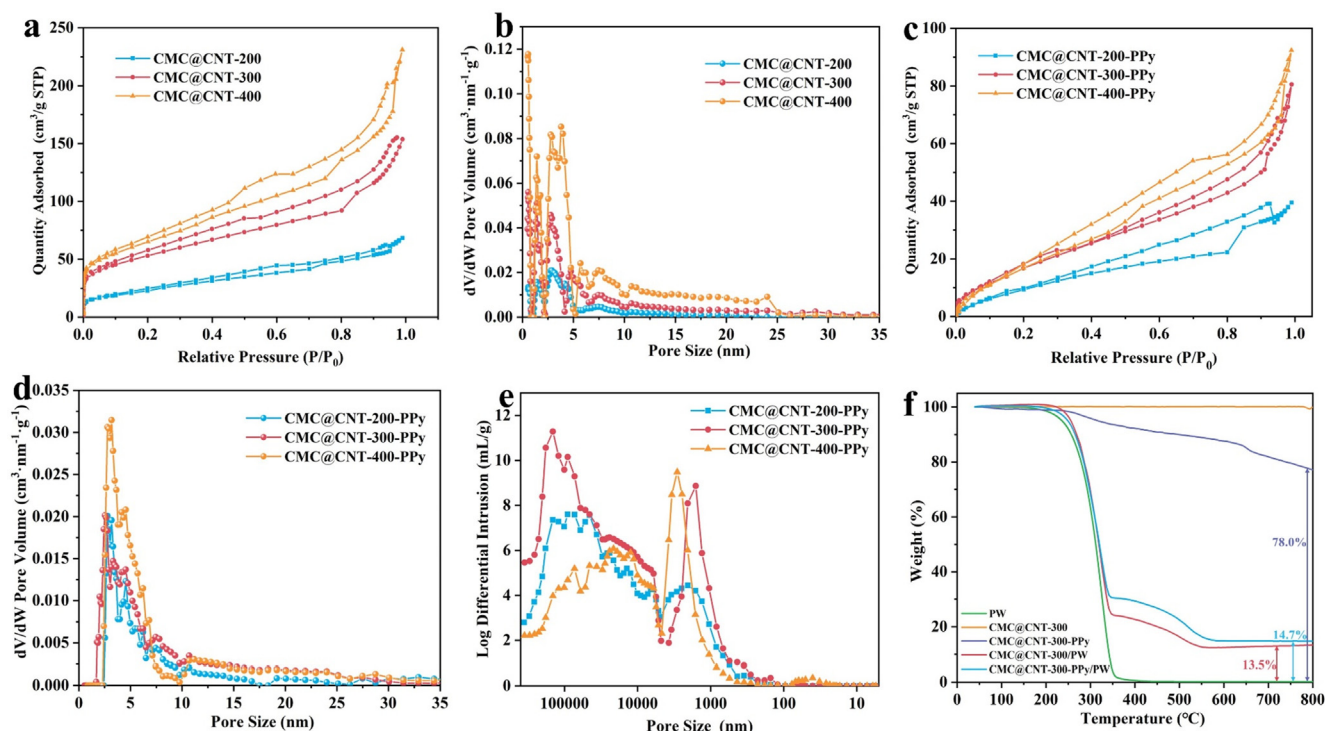


Fig. 4. Nitrogen adsorption and desorption curves of (a) CMC@CNT-X, (c) CMC@CNT-X-PPy and (b), (d) their corresponding pore size distribution; (e) Pore size distribution of CMC@CNT-X-PPy calculated by mercury extrusion curves; (f) TGA curves of the samples.

Table 1

Specific surface area and pore structure data of the aerogel supports calculated by nitrogen adsorption and desorption curves.

Samples	Specific Surface Area (m ² /g)	Pore Volume (cc/g)	Average Pore Size (nm)
CMC@CNT-200	87.7 ± 2.3	0.09 ± 0.01	1.9 ± 0.2
CMC@CNT-200-PPy	45.1 ± 1.2	0.06 ± 0.01	2.7 ± 0.1
CMC@CNT-300	189.6 ± 2.5	0.23 ± 0.02	0.6 ± 0.1
CMC@CNT-300-PPy	74.7 ± 1.3	0.11 ± 0.01	2.5 ± 0.2
CMC@CNT-400	234.6 ± 4.4	0.45 ± 0.02	0.6 ± 0.1
CMC@CNT-400-PPy	81.6 ± 3.2	0.15 ± 0.01	3.2 ± 0.1

The data calculated with three times tests.

change significantly after the polypyrrole coating is deposited. The porosity of aerogel supports can still reach about 85%, with only a slight decrease as the PPy coating formed. This is due to the thin deposition of polypyrrole on aerogel without occupying excessive macropore space, which will not change the original porous structure of the CNT aerogel support. Thus, the sufficient porosity of aerogel supports provides a guarantee for high loading amount of PCM and also high heat storage density of the composite PCMs.

Thermal stability of samples is analyzed by TGA curves (Fig. 4f). The aerogel supports of CMC@CNT-300 exhibits excellent thermal stability up to 800 °C. The CMC@CNT-300-PPy supports present continuous mass reduction starting at 300 °C and the remaining mass fraction at 800 °C is 78% due to the pyrolysis of PPy. The pristine PW shows completely thermal decomposition between 200 °C and 500 °C in a nitrogen atmosphere. The composite PCMs of CMC@CNT-300/PW and CMC@CNT-300-PPy/PW show a large mass loss at 250 °C to 650 °C, with a final residual mass of 13.5% and 14.7%, respectively. Obviously, the composite PCMs exhibit high thermal stabilities below 200 °C, which are vital for practical application. Depending on the remaining mass of each component, the loading mass fraction of PW in the CMC@CNT-300/PW and CMC@CNT-300-PPy/PW can be calculated as about 86.5% and 81.2%, respectively, which are almost consistent with the actual loading fraction of the corresponding samples (87.7 wt% and 83.2 wt%, obtained by experimental weighting). This indicates that the PPy coating does not significantly reduce the heat storage density of the composite PCMs.

3.2. Leakage-proof and thermophysical properties of composite PCMs

The shape-stability and leakage-proof properties of CMC@CNT-X-PPy/PW are verified by heating the samples on a platform at 65 °C, as shown in Figure S5. When the samples of CMC@CNT-X-PPy/PW (X = 200, 300 and 400, black disc) and PW (translucent disc) are placed on the heating platform, all samples are at room temperature and the shapes keep stable without significant changes. With the extension of the heating time, the temperature of the samples rise above the phase transition point (~60 °C) and the shapes of the composite PCMs remain stable, while the PW sample appears to have noticeable leakage. This is due to the adequate capillary and Van der Waals force provided by CMC@CNT-X-PPy porous aerogel supports effectively restrict the flow of liquid PW and prevent the leakage of PW. Thus, excellent shape-stability and leakage-proof properties of CMC@CNT-X-PPy/PW are proved directly which are beneficial to expanding the field of the practical application of the composite PCMs.

Thermal properties of the samples are exhibited by the DSC curves including the endothermic and exothermic processes (Fig. 5-a-5c). Compared with DSC curve of PW (Figure S6), CMC@CNT-X-PPy/PW samples present similar peaks. The weak peak in the low temperature range corresponds to the solid–solid phase transition, while the strong peak ascribes to solid–liquid transition of PCMs.

By further comparison, it is found that the DSC curves of CMC@CNT-X/PW are similar to CMC@CNT-X-PPy/PW, indicating that the deposition of PPy on the supports does not affect the melting and solidification process of the composite PCMs. The detailed data calculated according to the DSC curves are listed in Table 2. For the composite PCMs, the measured enthalpies during melting and freezing period are changed as the actual PW loading fraction of samples. After the deposition of PPy on the aerogels, the loading fractions of PW in CMC@CNT-X-PPy/PW only decrease slightly than before, and thus the latent heat values of CMC@CNT-X-PPy/PW are reduced by no more than 4% on average. Besides, the melting enthalpy changes of CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW are 160.9 J/g, 151.4 J/g, and 147.9 J/g, respectively, which are close to their theoretical enthalpy changes ($\Delta H_{\text{mt}} = 158.5 \text{ J/g}$, 154.1 J/g and 149.6 J/g respectively, calculated according to the actual PW loading of composite PCMs). These results suggest that the composite PCMs have high thermal storage capacity, and the deposition of PPy does not destroy the three-dimensional pore structure of the aerogel supports and has little influence on the thermal properties of the composite PCMs.

Reliability and stability of phase transition behavior are crucial for the sustainable application of PCMs. The DSC cyclic test of CMC@CNT-300-PPy/PW is conducted and the curves are almost coincident before and after 100 melting and freezing cycles (Fig. 5d). Only about a 3% decrease on average is detected of the enthalpy changes for the CMC@CNT-300-PPy/PW at the cyclic test, indicating that the CMC@CNT-300-PPy/PW has outstanding thermal stability. The excellent thermal cycling stability of CMC@CNT-300-PPy/PW should be assigned to the sufficient capillary force from CMC@CNT-300-PPy aerogel that effectively restricts the flow of the liquid phase and prevents leakage of PW.

In addition to thermal storage and stability, thermal conductivity is also a crucial property of composite PCMs. To investigate the effect of CMC@CNT-X aerogel network and PPy coating on the thermal transfer performance of composite PCMs, the thermal conductivities of PW and the samples are measured, and the value is $0.25 \text{ W/m} \cdot \text{K}^{-1}$, $0.37 \text{ W/m} \cdot \text{K}^{-1}$, $0.44 \text{ W/m} \cdot \text{K}^{-1}$, $0.41 \text{ W/m} \cdot \text{K}^{-1}$, $0.55 \text{ W/m} \cdot \text{K}^{-1}$, $0.51 \text{ W/m} \cdot \text{K}^{-1}$ and $0.64 \text{ W/m} \cdot \text{K}^{-1}$ (Fig. 5d). Obviously, all the thermal conductivities of composite PCMs have great improvement and the maximum increase of CMC@CNT-400-PPy/PW reaches 156% over pristine PW. In particular, under the same CNT content, the thermal conductivity of the composite PCMs with PPy coating is higher than that of the composite PCMs without PPy. The average increase of the thermal conductivity resulting from the PPy coating is about 25%. To verify the improvement in thermal transfer performance of CMC@CNT-X-PPy/PW composite PCMs, the behaviors during the thermal charging process are recorded by an infrared thermal imager. As shown in Fig. 5f, the samples with 1 mm thickness (from left to right represent PW, CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW) are placed in a heating platform at 40 °C. At first, all samples appear deep blue in the infrared image due to the upper surface temperatures are same as the ambient temperature (~15 °C). After constant heating for 15 s, the color of CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW changes to green, which means the surface temperature of the samples reach about 25 °C according to the temperature scale, while the PW still keep blue. The color of all the samples changed after 33 s, the CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW change to deep red (~35 °C), meanwhile, PW changes to yellow (~30 °C). With the extension of heating time, the color of CMC@CNT-400-PPy/PW changes to pink firstly (~40 °C), and the color of CMC@CNT-200-PPy/PW and CMC@CNT-300-PPy/PW samples is between pink and red, while the color of PW is still red at the moment. The color variation of the infrared image proves that

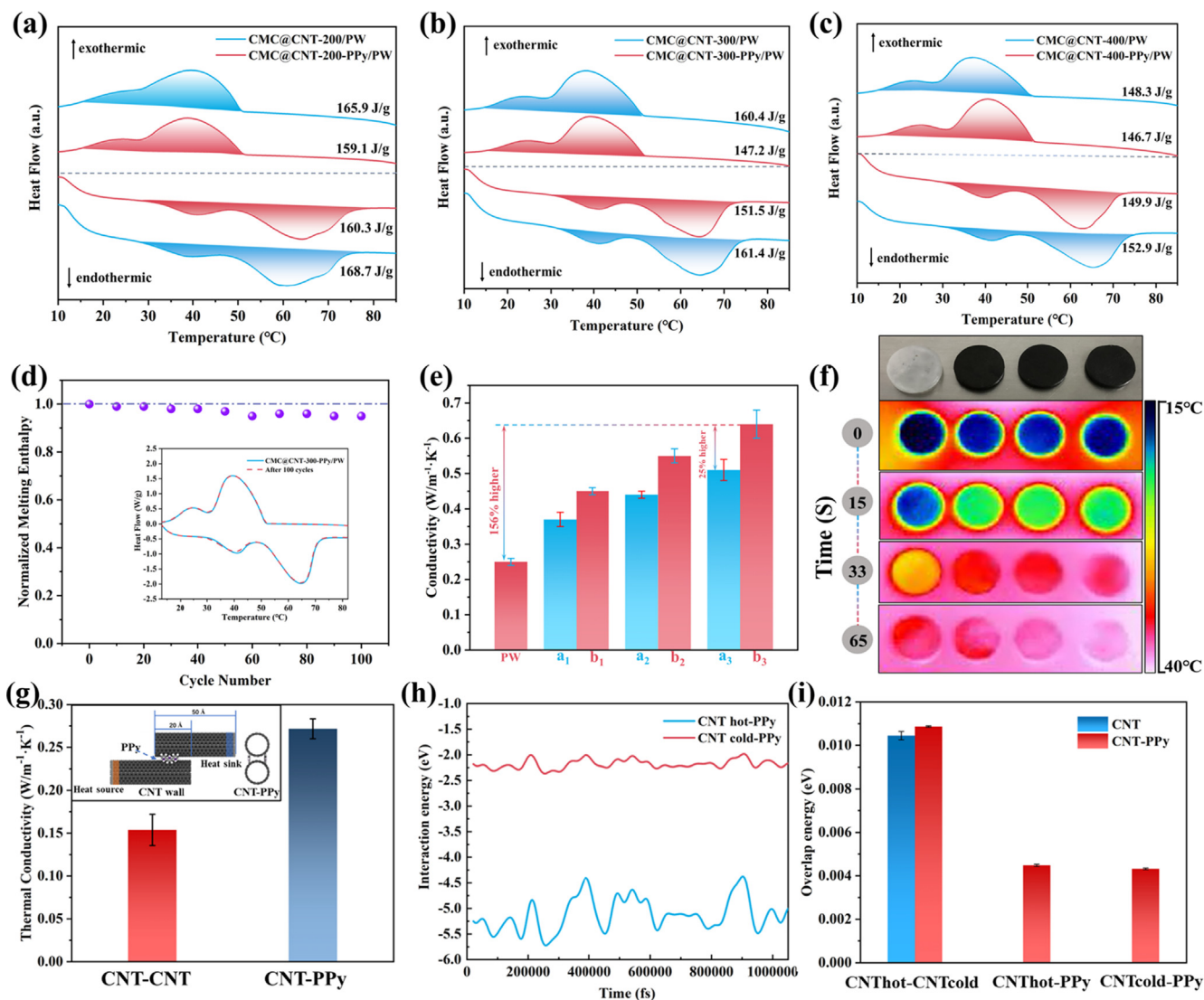


Fig. 5. (a–c) DSC curves of composite PCMs; (d) normalized melting enthalpy changes and DSC cycling curves of CMC@CNT-300-PPy/PW with 100 thermal cycles; (e) bar chart of thermal conductivity of samples (a_1 to a_3 represent CMC@CNT-200/PW, CMC@CNT-300/PW and CMC@CNT-400/PW, respectively, and b_1 to b_3 represent CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW, respectively); (f) digital photos and infrared thermal images of samples that are heated on a hot platform of 40 °C (The samples from left to right are PW, CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW); (g) bar chart of interfacial thermal conductivity after five molecular dynamics simulations (the inset is schematic of the model); (h) interaction energy between PPY and CNT at the interface; (i) the overlap energy after five molecular dynamics simulations between hot end and cold end of CNT atoms and PPY at the interface.

Table 2
Phase change parameters of PW and composite PCMs.

Materials	Loading fraction	Melting enthalpy ratio to PW	Melting		Solidifying	
			Peak T_m (°C)	ΔH_m (J/g)	Peak T_s (°C)	ΔH_s (J/g)
Pure PW	–	100%	62.0 ± 0.3	185.0 ± 0.5	45.5 ± 0.2	181.6 ± 0.6
CMC@CNT-200/PW	89.0 wt%	90.6%	58.5 ± 0.2	167.7 ± 1.2	40.4 ± 0.3	165.4 ± 1.2
CMC@CNT-200-PPy/PW	85.7 wt%	86.9%	63.0 ± 0.4	160.9 ± 0.5	40.4 ± 0.1	159.5 ± 0.8
CMC@CNT-300/PW	87.7 wt%	87.2%	63.8 ± 0.1	161.4 ± 0.4	38.7 ± 0.4	160.4 ± 1.2
CMC@CNT-300-PPy/PW	83.3 wt%	81.9%	63.4 ± 0.2	151.5 ± 1.1	39.8 ± 0.3	147.4 ± 0.7
CMC@CNT-400/PW	82.2 wt%	82.6%	64.4 ± 0.2	152.9 ± 0.4	37.5 ± 0.2	148.3 ± 2.0
CMC@CNT-400-PPy/PW	80.9 wt%	80.0%	61.9 ± 0.3	147.9 ± 1.3	41.2 ± 0.2	146.7 ± 1.2

The loading fraction is calculated by actual weighing according to equation 1 and the data is obtained after 3 times repeat experiments.

the higher the thermal conductivity of the sample is, the faster the upper surface temperature rises, which is consistent with the measured thermal conductivity of the samples.

For the CMC@CNT-X-PPy/PW samples, there are two main reasons for their significant increase in thermal conductivity: (1) the

three-dimensional continuous CNT aerogel framework of CMC@CNT-X provide a pathway for efficient heat transfer of the composite PCMs, which reduces the energy dissipation of phonon transfer along the CNT network and improves the thermal transfer efficiency; (2) the continuous PPy coating formed between CNT

acts as a repeater for heat transfer between interfaces, which reduces phonon scattering and interfacial thermal resistance and thus improves the thermal conductivity of the composite PCMs [38,39].

Molecular dynamics simulation is employed to explore the mechanism of heat transport enhancement further. Since the heat transfer of overlap between CNT determines the thermal conductivity of the three-dimensional CNT aerogels and their composite PCMs, the atomic model of intertube contact is established (inset of Fig. 5g) to explore the interfacial thermal conductance changes and heat transfer mechanism before and after loading PPy between CNT. The interfacial thermal conductance of CNT is simulated by non-equilibrium molecular dynamics, and the heat transfer mechanism is analyzed by atom group temperature, interaction energy and phonon overlap energy. The interfacial thermal conductance between CNTs before and after adding PPy molecules is shown in Fig. 5g, and the interfacial thermal conductance is improved by about 77% after adding PPy molecules. The temperatures of carbon atoms of hot CNT and cold CNT, and PPy molecules in the interface are shown in Figure S7, indicating that PPy between CNT acts as an inter-tube heat transfer repeater. Part of heat flow beyond the range of van der Waals force between CNT realizes inter-tube heat transfer through PPy molecules, and thus additional heat transfer channels to enhance heat transfer is provided by PPy molecules. The interaction energy between PPy molecules and CNT atoms at the interface over time is shown in Fig. 5h, which proves the role of PPy as a heat transfer repeater. Meanwhile, due to differences in atomic mass and vibration frequency, the interaction between PPy molecules and CNT atoms will cause interatomic resonance and leads to a vibration frequency change of CNT atoms. Therefore,

the change of vibration matching level between atoms before and after PPy deposition is further analyzed, as shown in Fig. 5i, PPy molecules have a high vibration matching level with CNT atoms, and the matching degree between hot and cold CNT atoms is improved after the addition of PPy, confirming that the positive resonance induction effect activated by heterogeneous PPy on CNT atoms not only successfully cancels out the heterogeneous phonon scattering but also further enhances intertube heat conduction. In brief, the deposition of PPy coating on the CNT aerogels effectively builds additional heat transfer channels and enhances the heat transfer between CNT by coordinating the carbon atom vibration. Consequently, the enhancement of heat transfer efficiency between tubes further promotes the heat transfer efficiency of three-dimensional porous CNT aerogels and thus improves the thermal conductivity of composite PCMs.

3.3. Solar-thermal energy conversion and storage of composite PCMs

In addition to improving heat transfer efficiency, the PPy coating deposited on CNT aerogels possesses strong light absorption and solar-thermal conversion capability. The light absorption capacity at a broadband from 200 to 2500 nm are measured by UV-vis-NIR spectrophotometer. As shown in Fig. 6a, CMC@CNT-300-PPy aerogel presents stronger broadband light absorption capacity than CMC@CNT-300 with an average increase of about 10%. The solar-thermal conversion and storage ability of PW, CMC@CNT-X/PW and CMC@CNT-X-PPy/PW are evaluated by simulating solar radiation. The schematic diagram of the measurement system is exhibited in Fig. 6b, the simulated light is provided by a xenon lamp equipped with an AM1.5 filter and the samples of the

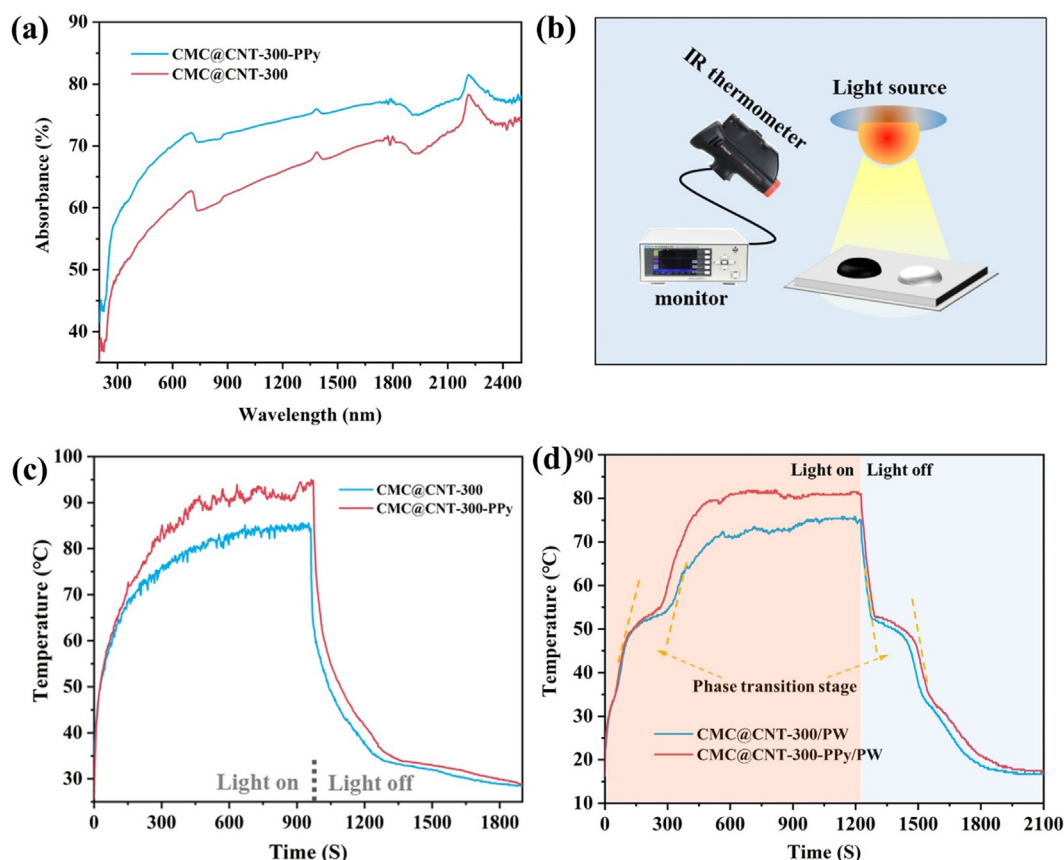


Fig. 6. (a) UV-vis-NIR absorption spectra of CMC@CNT-300 and CMC@CNT-300-PPy, (b) Schematic illustration of the photo-thermal performance measurement, the pellet samples are inserted on a thermally insulating polystyrene foam board; Time-temperature curves of (c) aerogel CMC@CNT-300, CMC@CNT-300-PPy and (d) composite PCMs CMC@CNT-300/PW, CMC@CNT-300-PPy/PW under simulated solar light irradiation (1 sun).

same size (1 mm in height, 12.7 mm in diameter) are prepared to ensure the same lighting area ($\sim 1.2 \text{ cm}^2$) [40]. The infrared thermometer is used to record the temperature of sample surface and the data are transmitted to the monitor in real time. The light absorption and solar-thermal conversion capability of CMC@CNT-300 and CMC@CNT-300-PPy aerogel supports are firstly measured under the light intensity of one sun ($\sim 1000 \text{ W/m}^2$), as the time-temperature curves presented in Fig. 6c. The surface temperature of CMC@CNT-300 and CMC@CNT-300-PPy samples are constantly increased from 20°C to about 60°C after irradiation for 100 s. Subsequently, the temperature of CMC@CNT-300 rises slowly and stabilizes at about 82°C , while the surface temperature of CMC@CNT-300-PPy sample can reach and stabilize at about 90°C under same illumination time. This may result from the deposition of PPy coating on the aerogel supports which enhances the light absorption and solar-thermal conversion capability of samples. After loading PW, the solar-thermal conversion and storage properties are detected of CMC@CNT-300/PW and CMC@CNT-300-PPy/PW samples and the corresponding time-temperature curves appear in Fig. 6d. The surface temperatures are constantly increased to about 50°C within 100 s for the two samples and then the temperature plateau emerges after the 150 s irradiation. As the phase transition is finished, the surface temperature of CMC@CNT-300/PW and CMC@CNT-300-PPy/PW continues to rise rapidly and then remain relatively stable at about 72°C and 81°C , respectively. The temperature difference is caused by the light absorption capacity and solar-thermal conversion efficiency of composite PCMs. Compared with CMC@CNT-300/PW sample, CNT and PPy heterostructure aerogel supports endows the composite PCMs with stronger light absorption and solar-thermal conversion characteristics. Upon the light is turned off, the surface temperature of samples drops

dramatically and a typical cooling plateau emerges at about 50°C , which is attributed to the phase transition from liquid to solid of the PCM. The emergence of a plateau demonstrates the effect of temperature control during the heating and cooling period. However, the pristine PW shows an unsatisfactory solar-thermal conversion capacity, and the maximum surface temperature only reaches 40°C during the irradiation time (Figure S8), which is even below the solid-liquid phase transition temperature point. Further, the solar-thermal energy storage efficiency (η_1) is calculated according to equation (2), including a series of composite PCMs before and after the PPy coating formed according to Figure S8, and the detailed parameters and the efficiency results of samples are listed in Table S2. Under the same simulated solar irradiation, the solar-thermal energy storage efficiency of CMC@CNT-X-PPy/PW samples is higher than that of CMC@CNT-X/PW at the phase transition period, and the maximum efficiency can reach 90% for CMC@CNT-400-PPy/PW. The results suggest that CNT and PPy heterogeneous porous aerogel supports can not only improve the heat transfer efficiency of composite PCMs but also improve their solar-thermal energy storage efficiency, which lays a foundation for the application of solar-responsive composite PCMs.

3.4. Electric-thermal energy conversion and storage of composite PCMs

Electro-thermal conversion performance of CMC@CNT-X-PPy/PW is investigated by a direct current (DC) power supply heating system, as shown in Fig. 7a. The sample of composite PCMs is attached with two copper sheet electrodes which are connected to a DC power supply, and the temperature evolution of the sample is measured by a fiber sensor (Fiso). The real-time temperature of samples is recorded and the temperature-time curves are exhib-

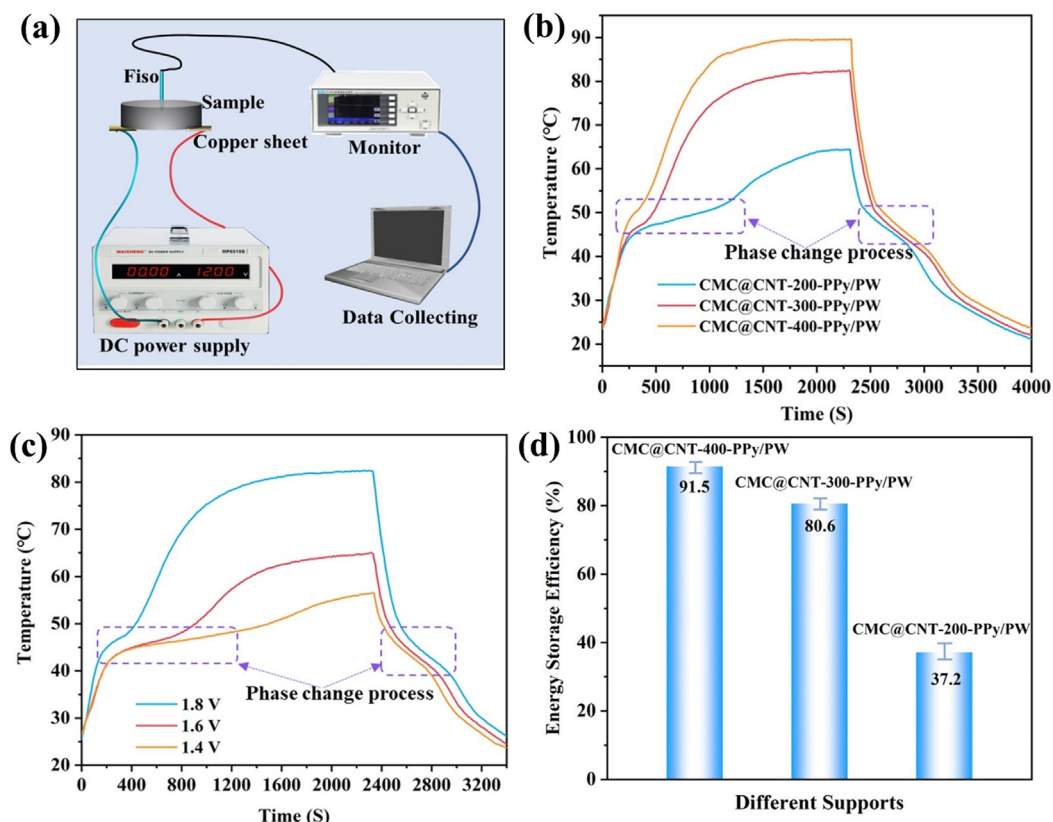


Fig. 7. (a) Schematic illustration of the electro-thermal conversion measurement, (b) Time-temperature curve of CMC@CNT-X-PPy/PW with an operating voltage of 1.8 V, (c) Time-temperature curve of CMC@CNT-300-PPy/PW with different operating voltage; (d) Bar diagram of electro-thermal energy storage efficiency of composite PCMs with different supports (Data calculated with three times tests).

ited in Fig. 7b. For the CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW samples, as the power supply is modulated to 1.8 V, the temperature of all the samples increases to about 42 °C constantly in 200 s. Next, a platform emerged from 43 °C to 55 °C that represents the latent thermal energy storage in the phase change process (the dotted area of Fig. 7b). Then, the temperature of CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW reaches about 65 °C, 82 °C, and 88 °C, respectively, and tends to be stable. Due to the different content of CNT, the joule heat emitted by CNT determines the heating rate and maximum temperature of composite PCMs under the operation voltage of 1.8 V. Meanwhile, the current value of CMC@CNT-200-PPy/PW, CMC@CNT-300-PPy/PW and CMC@CNT-400-PPy/PW is 445 mA, 551 mA and 630 mA, respectively, according to the power supply indication (Figure S9). Upon the power turn off, the temperature of the sample drops rapidly until a cooling platform appears. Latent heat is released to slow down the temperature declining rate at this liquid–solid phase transition area. These results prove that the electric energy can be converted into thermal energy and stored in the CMC@CNT-X-PPy/PW composite PCMs. Besides, the electro-thermal conversion property of CMC@CNT-X/PW samples are also shown in Figure S10, and it can be found that the introduction of PPy still maintains the efficient electric-thermal energy conversion and storage properties of composite PCMs. To explore the influence of operating voltage on the electro-thermal conversion performance of composite PCMs, the temperature–time curves of CMC@CNT-300-PPy/PW at different voltages are shown in Fig. 7c. The rate of temperature increase is proportional to the operating voltage of samples, and as the operating voltage is 1.4 V, the maximum temperature of CMC@CNT-300-PPy/PW can only reach about 55 °C, resulting in an incomplete phase transformation process. Therefore, an appropriate operating voltage plays a critical effect in the electro-thermal conversion and storage capacity of composite PCMs. Electro-thermal energy storage efficiency is also calculated according to Formula 3, and the results are presented in Fig. 7d. Electro-thermal energy storage efficiencies of the composite PCMs achieve 37.2%, 80.6%, and 91.5% when the aerogel supports are CMC@CNT-200-PPy, CMC@CNT-300-PPy and CMC@CNT-400-PPy, respectively, which are much higher than those in most previous work [35,41–43]. Hence, multiple energy conversion and storage functions are skillfully endowed to the PCMs can enrich the energy response and capture mode and significantly expand the application field of composite PCMs.

4. Conclusions

In summary, the shape stabilized composite PCMs of CMC@CNT-X-PPy/PW with enhanced thermal conductivity and efficient solar-/electro- thermal energy conversion and storage properties are facilely fabricated via introducing CNT and polypyrrole heterogeneous 3D porous aerogel supports with high thermal conductivity and high solar-thermal conversion efficiency. The results demonstrate that the composite PCMs of CMC@CNT-400-PPy/PW achieve an outstanding solar-/electro-thermal conversion efficiency, high thermal storage density (melting enthalpy changes of 147.9 J/g for the loading amount of 80.9 wt%) and excellent cycling thermal stability. In particular, the in-situ polymerized polypyrrole coating between carbon tubes can effectively improves the thermal conductivity of composite PCM (25% higher on average than that of the composite PCM without PPy coating) without decrease the heat storage density, which are significantly higher than the CNT-based composite PCMs reported in previous studies [44–46]. Molecular dynamics simulation is employed to explore the mechanism of heat transport enhancement and conclude that

the deposition of PPy coating on the CNT aerogels effectively builds additional thermal transfer channels and enhances the heat transport between CNT by coordinating the carbon atom vibration. Herein, the in-situ polymerization of conductive polymers on 3D networks can be used as an effective method to further improve the heat transport efficiency of porous network-based composite PCMs. The obtained composite PCMs with excellent comprehensive performance and solar/electrical multiple response characteristics can effectively broaden the application of PCM in thermal management.

Author contributions

The manuscript was written with the contributions of all authors. All authors have approved the final version of the manuscript.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2022.09.103>.

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